Fraud Detection With Graph Neural Networks: A Benchmark of SAGAD, SpaceGNN, and CRoC

COMP8851 Phase 2 Benchmark and GPU Compute Report

Overview

This report compares three recent graph neural networks for fraud detection: SAGAD (WWW 2026), SpaceGNN (ICLR 2025), and CRoC (ECAI 2025). We ran each one on six datasets (YelpChi, Amazon, T-Finance, Elliptic, FDCompCN, and T-Social) with three seeds (2, 42, and 72). That makes 18 model and dataset pairs and 54 final runs, and all of them finished. Our main metric is the area under the precision–recall curve (AUPRC). SpaceGNN scored highest on five of the six datasets. SAGAD scored highest on Elliptic and was also the fastest. Everything ran on one RTX A6000 (48 GB) for 116.11 billable hours, which cost $65.02 against a $65.00 budget.

Table 1

Test Performance, Training Speed, and Peak GPU Memory by Model and Dataset

Model	Dataset	AUPRC	AUROC	Macro-F1	Fraud F1	G-Mean	Epochs	s/epoch	Training time	Wall time	GPU (GiB)
SAGAD	YelpChi	.506 (.024)	.834 (.008)	.701 (.011)	.496 (.009)	.691 (.018)	100	0.0177	1.77s	89.0s (1.5m)	0.19
SAGAD	Amazon	.878 (.010)	.969 (.001)	.918 (.005)	.850 (.009)	.892 (.004)	100	0.0402	4.02s	50.33s	0.08
SAGAD	T-Finance	.812 (.060)	.940 (.021)	.618 (.495)	.573 (.420)	.585 (.467)	59.7	0.0416	2.48s	107.7s (1.8m)	0.15
SAGAD	Elliptic	**.703 (.007)**	.892 (.003)	.816 (.016)	.662 (.029)	.813 (.006)	98.3	0.0312	3.07s	87.4s (1.5m)	1.31
SAGAD	FDCompCN	.197 (.077)	.636 (.103)	.567 (.055)	.231 (.110)	.465 (.148)	73.7	0.0353	2.60s	30.83s	0.09
SAGAD	T-Social	.883 (.004)	.983 (.000)	.913 (.003)	.832 (.006)	.885 (.005)	100	0.1655	16.55s	13,720.1s (3.81h)	7.47
SpaceGNN	YelpChi	**.729 (.008)**	.912 (.006)	.804 (.004)	.661 (.004)	.770 (.014)	84.7	82.89	7,017.7s (1.95h)	7,281.3s (2.02h)	2.96
SpaceGNN	Amazon	**.915 (.001)**	.974 (.000)	.927 (.008)	.868 (.015)	.912 (.009)	61.3	11.27	691.1s (11.5m)	779.8s (13.0m)	9.49
SpaceGNN	T-Finance	**.905 (.006)**	.973 (.002)	.928 (.005)	.861 (.010)	.889 (.009)	71	53.47	3,796.6s (1.05h)	4,160.0s (1.16h)	**17.68**
SpaceGNN	Elliptic	.613 (.058)	.881 (.002)	.746 (.046)	.546 (.074)	.818 (.006)	40.7	36.60	1,488.5s (24.8m)	1,539.4s (25.7m)	0.25
SpaceGNN	FDCompCN	**.224 (.024)**	.645 (.024)	.572 (.006)	.225 (.013)	.433 (.020)	55	1.58	87.2s (1.5m)	110.5s (1.8m)	0.08
SpaceGNN	T-Social	**.962 (.012)**	.994 (.003)	.967 (.006)	.936 (.011)	.949 (.010)	2.3	8,107.44	18,917.4s (5.25h)	19,822.4s (5.51h)	5.98
CRoC	YelpChi	.584 (.076)	.870 (.034)	.742 (.035)	.565 (.053)	.744 (.021)	63.3	0.6438	40.78s	215.4s (3.6m)	3.34
CRoC	Amazon	.855 (.032)	.964 (.008)	.878 (.014)	.779 (.024)	.886 (.011)	100	0.3242	32.42s	484.8s (8.1m)	12.86
CRoC	T-Finance	.810 (.004)	.946 (.001)	.883 (.004)	.775 (.008)	.827 (.009)	100	0.5646	56.46s	930.1s (15.5m)	**19.69**
CRoC	Elliptic	.469 (.041)	.825 (.014)	.574 (.039)	.313 (.038)	.753 (.018)	96.3	0.5914	56.98s	220.6s (3.7m)	0.49
CRoC	FDCompCN	.109 (.003)	.520 (.009)	.490 (.016)	.086 (.075)	.245 (.214)	34	0.2448	8.32s	21.44s	0.12
CRoC	T-Social	.772 (.046)	.970 (.004)	.862 (.019)	.732 (.037)	.843 (.016)	27.7	43.00	1,189.5s (19.8m)	5,667.2s (1.57h)	4.27

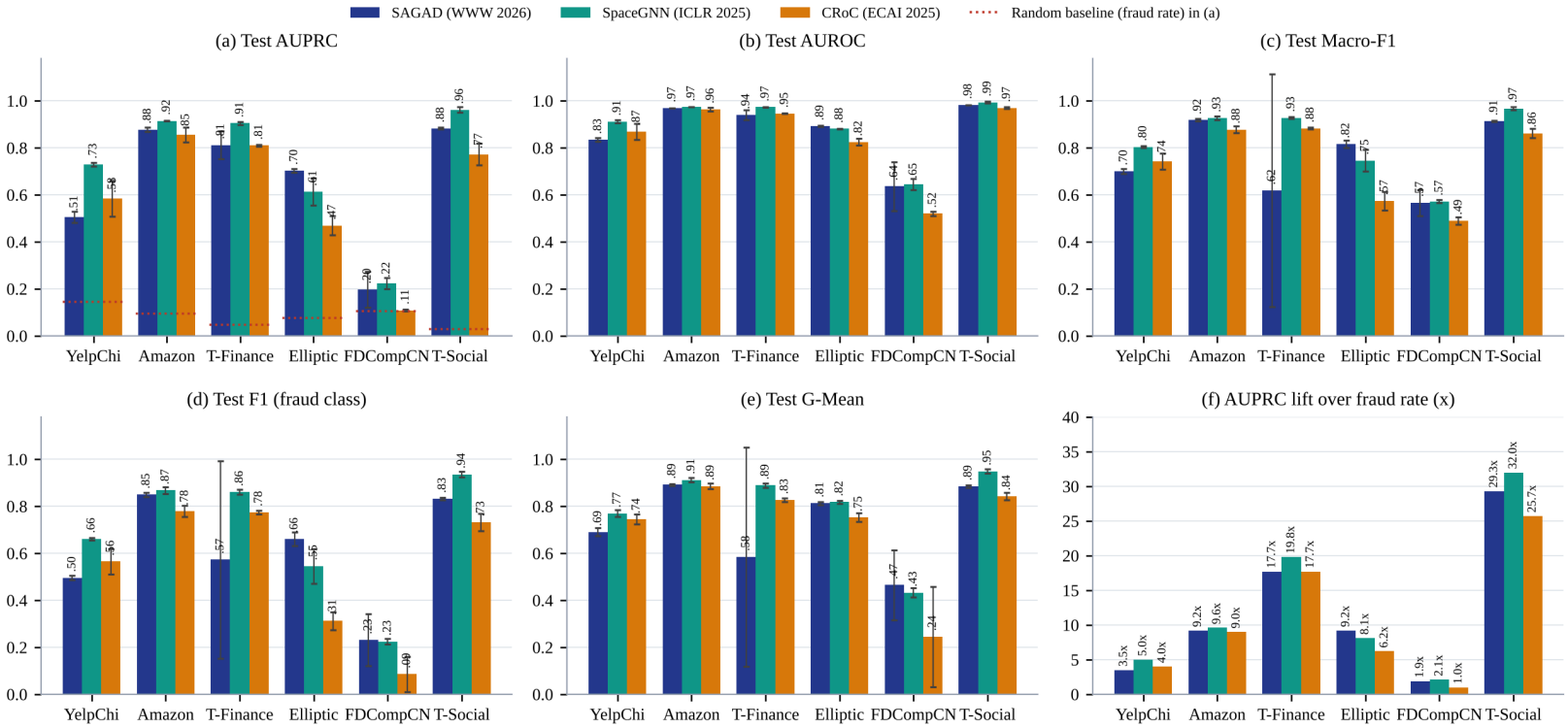
Note. Values are means with sample standard deviations in parentheses across seeds 2, 42, and 72. Bold marks the highest AUPRC on each dataset. Training time is seconds per epoch multiplied by epochs; wall time also includes graph loading, preprocessing, and validation. GPU = peak allocated memory.

Predictive Performance

Figure 1 shows how the three models did on five metrics, plus how much better each one is than random guessing (lift). SpaceGNN was ahead on most datasets, and its lead was biggest on the large graphs.

Figure 1

Test Performance of Each Model on Six Fraud Datasets



Note. Bars show means across three seeds and error bars show ±1 sample *SD*. Dotted red lines in Panel a mark the random-ranker baseline (the test fraud rate). Lift in Panel f is AUPRC divided by the fraud rate.

SpaceGNN maps multi-hop features into hyperbolic, spherical, and flat (Euclidean) spaces. It did best on the big, well-connected graphs: an AUPRC of .962 on T-Social (about 32 times the fraud rate), .915 on Amazon, .905 on T-Finance, and .729 on YelpChi.

SAGAD summarizes each node's neighborhood with quantile features at several scales, then uses Chebyshev spectral filters and gated fusion. It came first on Elliptic, the graph that is ordered by time (AUPRC = .703, Macro-F1 = .816), and scored .883 on T-Social.

CRoC trains with contrastive learning across two views of the graph and keeps a queue of 10,240 entries, which is meant to help against fraudsters who imitate normal users. Its best scores were .855 on Amazon, .810 on T-Finance (*SD* = .004), .772 on T-Social, and .584 on YelpChi.

Head-to-Head Comparison

Table 2 and Figure 2 put the three models side by side on test AUPRC. SpaceGNN was highest on YelpChi (.729), Amazon (.915), T-Finance (.905), FDCompCN (.224), and T-Social (.962). SAGAD was highest on Elliptic (.703), ahead of SpaceGNN (.613) and CRoC (.469). CRoC was second on YelpChi (.584) and came within .002 of SAGAD on T-Finance (.810 vs. .812). All three models beat the random ranker everywhere except CRoC on FDCompCN, where it was only just above chance (.109 vs. .105).

Table 2

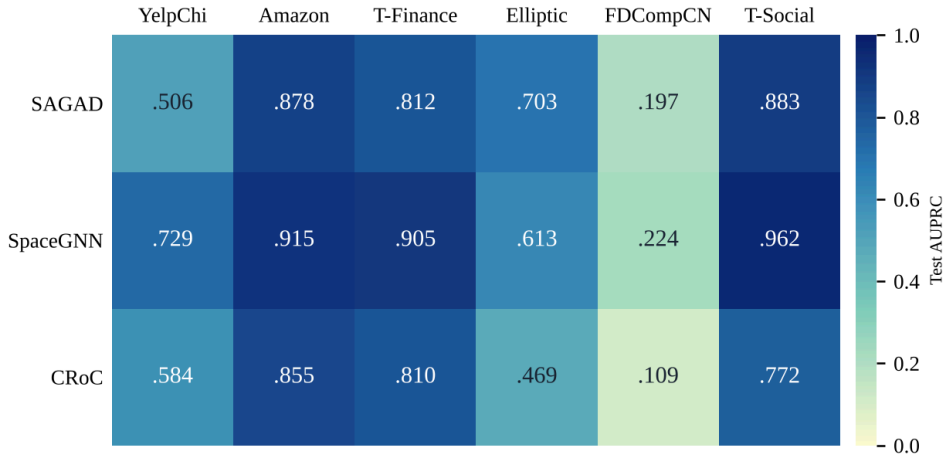
Test AUPRC for SAGAD, SpaceGNN, and CRoC on All Six Datasets

Model	YelpChi	Amazon	T-Finance	Elliptic	FDCompCN	T-Social
SAGAD	.506 (.024)	.878 (.010)	.812 (.060)	**.703 (.007)**	.197 (.077)	.883 (.004)
SpaceGNN	**.729 (.008)**	**.915 (.001)**	**.905 (.006)**	.613 (.058)	**.224 (.024)**	**.962 (.012)**
CRoC	.584 (.076)	.855 (.032)	.810 (.004)	.469 (.041)	.109 (.003)	.772 (.046)
Random ranker	.145	.095	.046	.076	.105	.030

Note. Values are means with sample standard deviations in parentheses across seeds 2, 42, and 72. Bold marks the highest AUPRC on each dataset. The random ranker row gives the fraud rate in the test set, which is the AUPRC you would expect from random scores.

Figure 2

Test AUPRC Heatmap for the Three Models

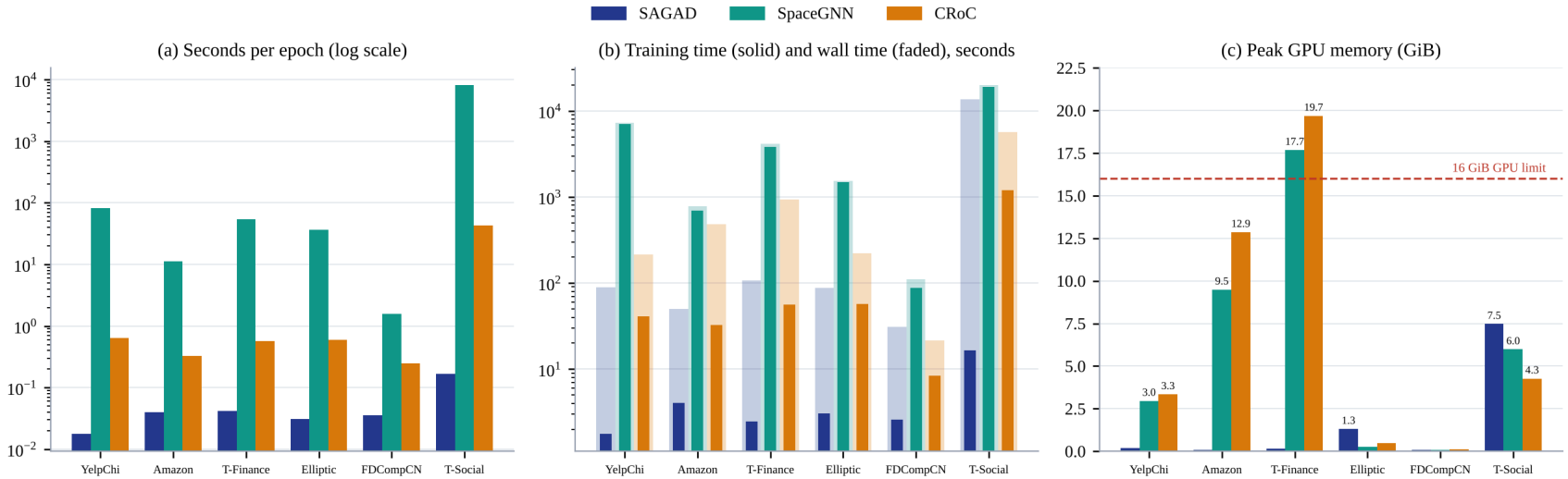


Note. Darker cells indicate higher AUPRC.

Speed and Memory

Figure 3

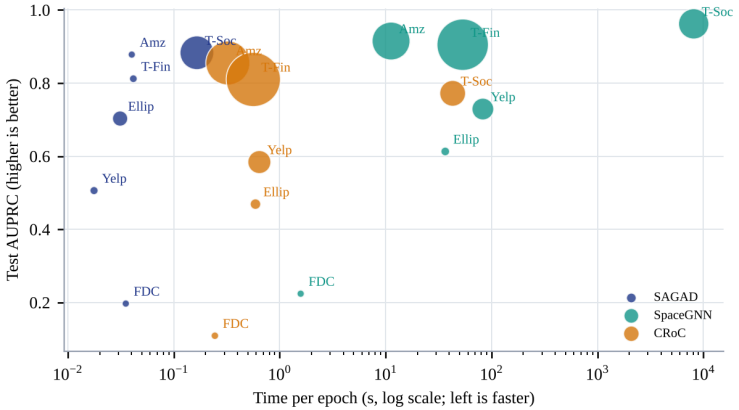
Training Speed, Total Time, and Peak GPU Memory



Note. Panel b shows training time (seconds per epoch × epochs) as solid bars and total wall-clock time as faded bars. The dashed line in Panel c is the 16 GiB limit of a GPU such as the Tesla T4.

Figure 4

Test AUPRC Against Time per Epoch



Note. Bubble area is proportional to peak GPU memory.

Training time and preprocessing. SAGAD builds its multi-hop quantile features once, before the first epoch. After that, training takes .0177 to .0416 s per epoch on the five smaller graphs and .1655 s per epoch on T-Social. Most of its 3.81 hr of wall time on T-Social goes to extracting 3-hop features across 73.1 million edges.

SpaceGNN on T-Social. SpaceGNN works through three curvature spaces on mini-batches of 50 nodes (46,249 batches per epoch), so one epoch takes about 8,107 s (2.25 hr). The runs averaged 2.3 epochs, for 18,917 s (5.25 hr) in total, which stayed inside the 360-min cap. It still reached an AUPRC of .962.

GPU memory. On T-Finance, which has 21.2 million edges, SpaceGNN used 17.68 GiB and CRoC used 19.69 GiB. A 16 GiB card such as the Tesla T4 cannot hold that, so we needed a larger GPU to finish every run.

GPU Budget and Timeline

Every run used one Vast.ai RTX A6000 (48 GB) at $0.56 per hour. We were billed for 116.11 hr (4.84 days, October 11 to 16, 2026), which came to $65.02, or $0.02 over the $65.00 budget. Running two workers at the same time fit 172.51 hr of job compute into those 116.11 billable hours and saved about $31.58.

Table 3

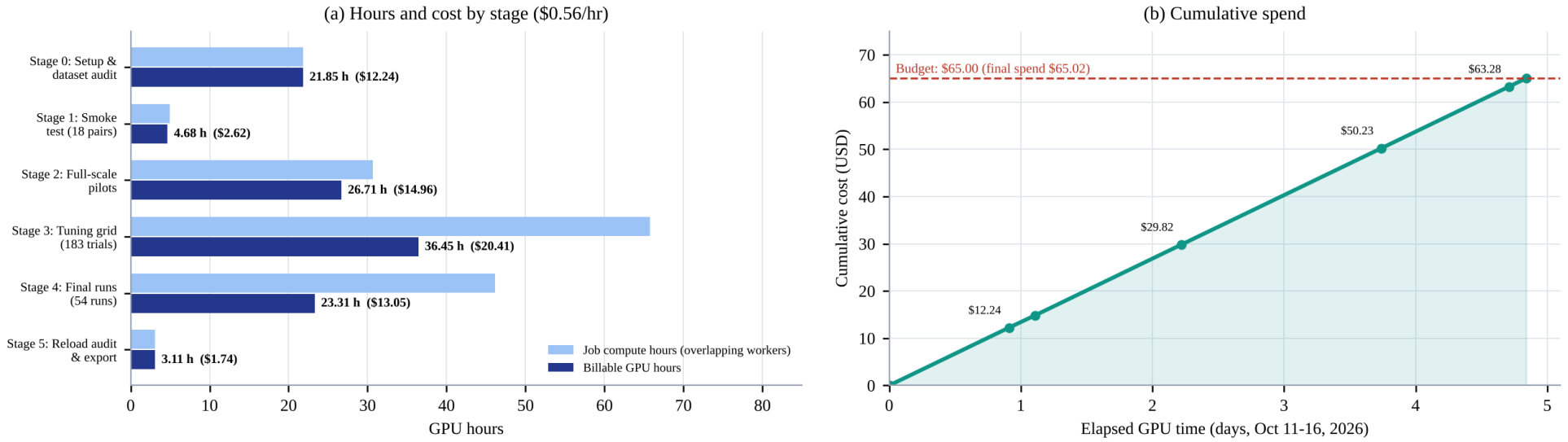
GPU Runtime and Cost by Stage

Stage	Activity	Dates (2026)	Job compute (hr)	Billable (hr)	Cost
0	Setup: instance, CUDA 12.1 and DGL 2.4 build, dataset upload, split audit	Oct 11–12	21.85	21.85	$12.24
1	Smoke test of all 18 pairs (2 epochs each)	Oct 12	4.94	4.68	$2.62
2	Full-scale pilots (FDCompCN and T-Social, including a 24.34 hr SpaceGNN pilot)	Oct 12–13	30.66	26.71	$14.96
3	Hyperparameter tuning (183 trials, two workers)	Oct 13–15	65.78	36.45	$20.41
4	Final runs: 18 pairs × 3 seeds, two workers	Oct 14–16	46.17	23.31	$13.05
5	Reload check of 54 checkpoints, metric recomputation, archive export	Oct 16	3.11	3.11	$1.74
Total	**Full benchmark on one RTX A6000 (48 GB)**	**Oct 11–16**	**172.51**	**116.11**	**$65.02**

Note. Cost is billable hours at $0.56 per hr. The stages cover 261 GPU runs in all (18 smoke tests, 6 pilots, 183 tuning trials, and 54 final runs), each timestamped in the ledger file. Job compute is higher than billable time wherever two workers ran at once.

Figure 5

Billable GPU Hours and Cumulative Spend



Datasets, Hyperparameters, and Protocol

Table 4

Dataset Characteristics

Dataset	Nodes	Labeled nodes	Edges	Features	Relations	Train / validation / test nodes	Test fraud rate (%)	Fraud-node homophily
YelpChi	45,954	45,954	3,846,979	32	3	18,381 / 9,191 / 18,382	14.53	.805
Amazon	11,944	8,639	4,417,576	25	3	3,455 / 1,728 / 3,456	9.52	.862
T-Finance	39,357	39,357	21,222,543	10	1	15,742 / 7,871 / 15,744	4.58	.456
Elliptic	203,769	46,564	234,355	166	1	18,625 / 9,314 / 18,625	7.61	.490
FDCompCN	5,317	5,317	3,634	57	3	2,126 / 1,063 / 2,128	10.48	.862
T-Social	5,781,065	5,781,065	73,105,508	10	1	2,312,426 / 1,156,213 / 2,312,426	3.01	.826

Note. Statistics are for the fixed 40/20/40 train, validation, and test split (split seed 2). Edges are undirected. Elliptic is directed and covers 49 time steps. Relations are the edge types in each graph.

Table 5

Tuned Hyperparameters for SAGAD and SpaceGNN

Dataset	SAGAD	SpaceGNN
YelpChi	K = 2, p = [.25, 1.0], hidden = [128, 128], lr = .01, dropout = .1, weight decay = 0, NodeDimGated(2)	6 layers, hidden = 128, lr = .001, dropout = .10, α = 1.0, β = 1.0, batch = 50, std = official
Amazon	K = 5, p = [.1, .8], hidden = [32], lr = .01, dropout = 0, weight decay = 0, Identity, NodeDimGated(1)	3 layers, hidden = 128, lr = .001, dropout = .05, α = 0, β = 0, batch = 50, std = official
T-Finance	K = 5, p = [1.0, 1.0], hidden = [128, 128, 128], lr = .1, dropout = .1, weight decay = 1e-4, feat = l1	3 layers, hidden = 128, lr = .001, dropout = .10, α = 1.0, β = 1.0, batch = 50, std = official
Elliptic	K = 4, p = [.4, 1.0], hidden = [128, 128, 128], lr = .01, dropout = .5, weight decay = 1e-4, Gate(1)	3 layers, hidden = 32, lr = .001, dropout = .05, α = 1.0, β = 1.0, batch = 50, std = .015
FDCompCN	K = 5, p = [.8, 1.0], hidden = [128], lr = .1, dropout = .2, weight decay = 0, NodeDimGated(2)	1 layer, hidden = 64, lr = .001, dropout = 0, α = 0, β = .5, batch = 50, std = .015
T-Social	K = 3, p = [0, 1.0], hidden = [128, 128], lr = .01, batch norm, feat = l1, Gate(2)	6 layers, hidden = 128, lr = .001, dropout = .05, α = .5, β = 1.0, batch = 50, std = official

Note. We chose all settings using validation AUPRC only. For CRoC, YelpChi, Amazon, and FDCompCN used CRoCGIN (2 layers, hidden = 64, lr = .003, batch = 1024, α = .75, η = .2, γ = .5, queue = 10,240); T-Finance used SAGE (α = .9, η = .8, γ = 1); Elliptic used SAGE (α = .2, η = .2, γ = 1); and T-Social used GIN (α = 0, η = .5, γ = 1).

Protocol. Each run trained for up to 100 epochs and stopped early after 20 epochs with no gain in validation AUPRC. We swept the decision threshold from .01 to .99, kept the one with the best validation Macro-F1, and applied it once to the test nodes. The unlabeled background nodes in Amazon (3,305) and Elliptic (157,205) were used only to pass messages, and validation and test labels were hidden before the model was set up. All 54 final runs saved their predictions, summary, and weights, and all 54 passed an independent reload check.

Appendix

Per-Seed Benchmark Results

Table A1

Results for Individual Runs 1–27 (SAGAD, and SpaceGNN on YelpChi, Amazon, and T-Finance)

#	Model	Dataset	Seed	Best / run epochs	Val. thr.	Val. AUPRC	Test AUPRC	AUROC	Macro-F1	Fraud F1	G-Mean	s/epoch	Training time	Wall time	GPU (GiB)
1	SAGAD	YelpChi	2	99/100	.62	.4797	.4940	.8278	.6985	.4941	.6967	0.0160	1.60s	91.4s (1.5m)	0.19
2	SAGAD	YelpChi	42	99/100	.61	.4830	.4892	.8299	.6910	.4876	.7054	0.0183	1.83s	93.4s (1.6m)	0.19
3	SAGAD	YelpChi	72	100/100	.68	.5233	.5334	.8428	.7121	.5050	.6716	0.0189	1.89s	82.3s (1.4m)	0.19
4	SAGAD	Amazon	2	100/100	.87	.8388	.8796	.9701	.9167	.8484	.8901	0.0394	3.94s	50.14s	0.08
5	SAGAD	Amazon	42	94/100	.85	.8654	.8878	.9682	.9230	.8599	.8973	0.0375	3.75s	50.37s	0.08
6	SAGAD	Amazon	72	100/100	.85	.8339	.8672	.9675	.9137	.8429	.8895	0.0437	4.37s	50.48s	0.08
7	SAGAD	T-Finance	2	94/100	.89	.8692	.8576	.9581	.9070	.8217	.8563	0.0428	4.28s	158.8s (2.6m)	0.15
8	SAGAD	T-Finance	42	38/58	.61	.8557	.8350	.9441	.9003	.8089	.8534	0.0393	2.28s	100.6s (1.7m)	0.15
9	SAGAD	T-Finance	72	1/21	.99	.7682	.7442	.9163	.0459	.0877	.0454	0.0421	0.88s	63.7s (1.1m)	0.15
10	SAGAD	Elliptic	2	90/100	.99	.7477	.7043	.8953	.8041	.6396	.8058	0.0309	3.09s	86.5s (1.4m)	1.31
11	SAGAD	Elliptic	42	75/95	.99	.7415	.6952	.8925	.8349	.6946	.8172	0.0316	3.00s	84.7s (1.4m)	1.31
12	SAGAD	Elliptic	72	90/100	.98	.7400	.7090	.8896	.8104	.6514	.8163	0.0311	3.11s	91.1s (1.5m)	1.31
13	SAGAD	FDCompCN	2	86/100	.60	.2999	.2364	.6835	.5986	.2826	.5118	0.0400	4.00s	43.65s	0.09
14	SAGAD	FDCompCN	42	1/21	.99	.1483	.1088	.5187	.5031	.1045	.2992	0.0411	0.86s	10.30s	0.09
15	SAGAD	FDCompCN	72	100/100	.54	.3206	.2464	.7070	.5984	.3056	.5829	0.0295	2.95s	38.53s	0.09
16	SAGAD	T-Social	2	100/100	.96	.8801	.8802	.9825	.9102	.8255	.8791	0.1643	16.43s	13,828.4s (3.84h)	7.47
17	SAGAD	T-Social	42	100/100	.95	.8817	.8804	.9828	.9140	.8330	.8886	0.1641	16.41s	13,762.4s (3.82h)	7.47
18	SAGAD	T-Social	72	99/100	.95	.8887	.8880	.9823	.9160	.8368	.8876	0.1683	16.83s	13,569.6s (3.77h)	7.47
19	SpaceGNN	YelpChi	2	66/86	.63	.7528	.7346	.9174	.8074	.6660	.7661	71.00	6,106.0s (1.70h)	6,353.3s (1.76h)	2.96
20	SpaceGNN	YelpChi	42	65/85	.70	.7374	.7321	.9131	.8042	.6599	.7588	72.58	6,169.1s (1.71h)	6,414.3s (1.78h)	2.96
21	SpaceGNN	YelpChi	72	63/83	.44	.7423	.7204	.9064	.7995	.6573	.7865	105.76	8,777.9s (2.44h)	9,076.4s (2.52h)	2.96
22	SpaceGNN	Amazon	2	66/86	.23	.8950	.9152	.9746	.9363	.8844	.9233	8.68	746.3s (12.4m)	849.4s (14.2m)	9.49
23	SpaceGNN	Amazon	42	36/56	.41	.8932	.9159	.9739	.9259	.8653	.9067	13.07	731.7s (12.2m)	820.7s (13.7m)	9.49
24	SpaceGNN	Amazon	72	22/42	.40	.8887	.9149	.9739	.9200	.8549	.9070	14.18	595.4s (9.9m)	669.4s (11.2m)	9.49
25	SpaceGNN	T-Finance	2	40/60	.65	.9206	.8984	.9715	.9223	.8510	.8780	59.83	3,589.7s (59.8m)	3,938.7s (1.09h)	17.67
26	SpaceGNN	T-Finance	42	48/68	.55	.9243	.9095	.9748	.9321	.8700	.8952	59.44	4,041.7s (1.12h)	4,428.5s (1.23h)	17.68
27	SpaceGNN	T-Finance	72	65/85	.49	.9229	.9078	.9727	.9281	.8623	.8926	44.21	3,758.3s (1.04h)	4,113.0s (1.14h)	17.68

Note. Each row is one run. Training time equals s/epoch multiplied by the number of epochs run. Val. thr. = validation-selected decision threshold.

Table A2

Results for Individual Runs 28–54 (SpaceGNN on Elliptic, FDCompCN, and T-Social, and CRoC)

#	Model	Dataset	Seed	Best / run epochs	Val. thr.	Val. AUPRC	Test AUPRC	AUROC	Macro-F1	Fraud F1	G-Mean	s/epoch	Training time	Wall time	GPU (GiB)
28	SpaceGNN	Elliptic	2	34/54	.99	.7997	.6029	.8832	.7016	.4760	.8199	36.41	1,966.2s (32.8m)	2,032.6s (33.9m)	0.25
29	SpaceGNN	Elliptic	42	7/27	.83	.7538	.6760	.8795	.7942	.6230	.8115	36.91	996.6s (16.6m)	1,030.8s (17.2m)	0.25
30	SpaceGNN	Elliptic	72	21/41	.99	.7606	.5616	.8809	.7426	.5387	.8223	36.65	1,502.7s (25.0m)	1,554.9s (25.9m)	0.25
31	SpaceGNN	FDCompCN	2	14/34	.30	.2930	.2048	.6278	.5657	.2108	.4145	1.93	65.7s (1.1m)	81.1s (1.4m)	0.08
32	SpaceGNN	FDCompCN	42	95/100	.23	.2996	.2518	.6727	.5764	.2292	.4303	1.52	152.0s (2.5m)	193.4s (3.2m)	0.08
33	SpaceGNN	FDCompCN	72	11/31	.13	.2675	.2168	.6355	.5748	.2347	.4541	1.41	43.69s	56.86s	0.08
34	SpaceGNN	T-Social	2	3/3	.73	.9688	.9692	.9949	.9706	.9428	.9537	7,984.11	23,952.3s (6.65h)	24,954.5s (6.93h)	5.98
35	SpaceGNN	T-Social	42	3/3	.21	.9694	.9695	.9952	.9699	.9415	.9545	8,256.47	24,769.4s (6.88h)	25,778.1s (7.16h)	5.98
36	SpaceGNN	T-Social	72	1/1	.22	.9493	.9486	.9906	.9605	.9232	.9373	8,030.29	8,030.3s (2.23h)	8,734.6s (2.43h)	5.98
37	CRoC	YelpChi	2	44/64	.58	.5966	.6157	.8832	.7615	.5950	.7552	0.6332	40.53s	217.8s (3.6m)	3.34
38	CRoC	YelpChi	42	85/100	.74	.6229	.6401	.8949	.7627	.5969	.7557	0.6514	65.1s (1.1m)	332.8s (5.5m)	3.34
39	CRoC	YelpChi	72	6/26	.55	.4728	.4974	.8314	.7008	.5043	.7197	0.6408	16.66s	95.5s (1.6m)	3.34
40	CRoC	Amazon	2	100/100	.92	.8336	.8650	.9635	.8790	.7813	.8766	0.3212	32.12s	479.6s (8.0m)	12.86
41	CRoC	Amazon	42	100/100	.68	.7957	.8203	.9558	.8634	.7546	.8832	0.3178	31.78s	485.7s (8.1m)	12.86
42	CRoC	Amazon	72	100/100	.79	.8401	.8812	.9718	.8904	.8024	.8980	0.3336	33.36s	489.0s (8.2m)	12.86
43	CRoC	T-Finance	2	99/100	.97	.8175	.8136	.9478	.8845	.7785	.8302	0.5532	55.32s	904.0s (15.1m)	19.69
44	CRoC	T-Finance	42	100/100	.92	.8136	.8065	.9450	.8780	.7657	.8161	0.5709	57.09s	913.5s (15.2m)	19.69
45	CRoC	T-Finance	72	99/100	.95	.8171	.8094	.9465	.8855	.7804	.8334	0.5698	56.98s	972.8s (16.2m)	19.69
46	CRoC	Elliptic	2	84/100	.83	.8653	.4878	.8408	.5752	.3116	.7561	0.5764	57.64s	226.0s (3.8m)	0.49
47	CRoC	Elliptic	42	74/94	.86	.8595	.4213	.8144	.6127	.3505	.7685	0.6106	57.39s	218.6s (3.6m)	0.49
48	CRoC	Elliptic	72	75/95	.75	.8145	.4976	.8189	.5339	.2754	.7331	0.5884	55.90s	217.2s (3.6m)	0.49
49	CRoC	FDCompCN	2	25/45	.50	.1972	.1099	.5290	.5008	.1412	.3946	0.2474	11.13s	27.65s	0.12
50	CRoC	FDCompCN	42	16/36	.51	.1870	.1123	.5203	.4974	.1165	.3406	0.2449	8.82s	22.80s	0.12
51	CRoC	FDCompCN	72	1/21	.50	.1635	.1061	.5102	.4724	.0000	.0000	0.2389	5.02s	13.85s	0.12
52	CRoC	T-Social	2	8/28	.90	.8234	.8229	.9730	.8826	.7719	.8554	41.50	1,161.9s (19.4m)	5,523.5s (1.53h)	4.27
53	CRoC	T-Social	42	7/27	.84	.7615	.7594	.9720	.8591	.7268	.8490	46.24	1,248.4s (20.8m)	5,974.0s (1.66h)	4.27
54	CRoC	T-Social	72	8/28	.82	.7337	.7344	.9664	.8448	.6987	.8255	41.37	1,158.4s (19.3m)	5,504.2s (1.53h)	4.27

Note. Environment: PyTorch 2.4.0+cu121, DGL 2.4.0+cu121, NVIDIA RTX A6000 (48,670 MB). Upstream commits: SAGAD faddcbca117b, SpaceGNN 921c03ff879b, CRoC 70cb7c6c654f. Raw predictions, aggregated tables, source code, and ledger logs are in the deliverable package.